

Finding a Relationship Between Circuit Resistance and its Effect on Sound Intensity of a Speaker

Introduction

After having been to several schools in Calgary, I have often noticed that the audible effect of the PA (Public Address) systems are inconsistent. They sound quite different from school to school, and sometimes even room to room: loud and audible in one room but barely audible in another. These large discrepancies I noticed in sound output intrigued me, and I wanted to further investigate the topic. As the purpose of PA systems in schools is to communicate to everyone in the building, making it loud and clearly audible is of utmost importance. On the other hand, keeping the sound output under regulatory sound levels is also important for the safety of the students and staff in the building—the safe hearing range for humans is sound levels at or lower than 70 decibels (*Keep Listening | What Are Safe Decibels? — Hearing Health Foundation*, n.d.). From these considerations, I began to ponder potential methods to mitigate an overly loud speaker sound output to regulate its levels and what variables could affect the sound level output. When I learned about circuits in school, I thought that one plausible solution could be the addition of resistors to the same circuit loop as the speakers. I thus want to investigate the relationship between the resistance in a circuit and the impact it has on the sound intensity of a speaker in this experiment to test the validity of my proposed solution.

Background

Ohm's Law states that the electrical current passing through two points in a circuit is in direct proportion to its voltage in those two points. It is mathematically given by the formula $V = IR$, where:

- V is the voltage, measured in volts (V)
- I is the current passing through the points, measured in amperes (A)
- R is the resistance in the circuit, measured in ohms (Ω)

It is followed by resistors wired in electrical circuits. These such resistors are called Ohmic resistors as they obey Ohm's Law, and cause voltage drops (*Fundamentals of Electricity*, n.d.) throughout a series circuit. Larger resistors will cause larger voltage drops throughout a series circuit, and thus diminish the amount of voltage in circuit components that are connected after the resistor in the circuit.

Sound level, measured in decibels (dB), is the logarithmic unit for measuring sound intensity. We perceive various sound levels from a day-to-day basis, from whispering sounds to the sirens on emergency assistance vehicles. The average sound level range of standard school PA systems is around 66 dB - 70 dB (Mesnik, 2014), making it 6 dB - 10 dB louder than average background noise (~60dB) in order to be heard over any classroom or background noise.

Electric power, measured in Watts (W), is the amount of work done in an electrical circuit in a given amount of time. It is mathematically given by the formula:

$$P = IV$$

Where:

- P is the electric power in the circuit, measured in Watts (W)
- I is the electric current flowing through the circuit, measured in amperes (A)
- V is the voltage in the circuit, measured in volts (V)

Sound intensity, I , measured in W/m^2 , is the amount of transmitting power a speaker has per unit area. The formula for sound level is given by:

$$L = 10 \log\left(\frac{I}{I_0}\right)$$

Where:

- L is the sound level of the source, measured in decibels (dB)
- I_0 is the reference intensity at the threshold of human hearing, 10^{-12} W/m^2 (*Sound Intensity Terms and Definitions*, n.d.)
- I is the sound intensity of the source, measured in W/m^2

Rearranging the formula for sound intensity, we get:

$$I = 10^{\frac{L}{10}} \cdot I_0$$

From this formula, as sound level increases, sound intensity exponentially increases with it. Thus, the impact of resistance in terms of whether it increases or decreases the sound level in this experiment will have the same impact on sound intensity.

The formula for sound intensity can also be written as:

$$I = \frac{P}{A}$$

Where:

- P is the electrical power, measured in Watts (W)
- A is the area, measured in metres squared (m^2)

Substituting the given formula for power, we obtain:

$$I = \frac{I_c V}{A}$$

Where:

- I_c is the current in the circuit (whereas I denotes the sound intensity)

In a given electrical circuit, sound intensity, I , varies directly with voltage, V , when current I_c and area A stay constant. This can be denoted by: $I \propto V$.

Speakers produce sound by converting electrical energy into mechanical energy.

Electric currents powering the speaker creates a magnetic field, causing movement in a coil and diaphragm inside it, which subsequently pushes surrounding air to produce sound that reaches our ears. (*How Do Speakers Work: A Guide on How Speakers Produce Sound*, 2024)

Design:

Figure 1: A schematic diagram of the circuit configuration for this lab

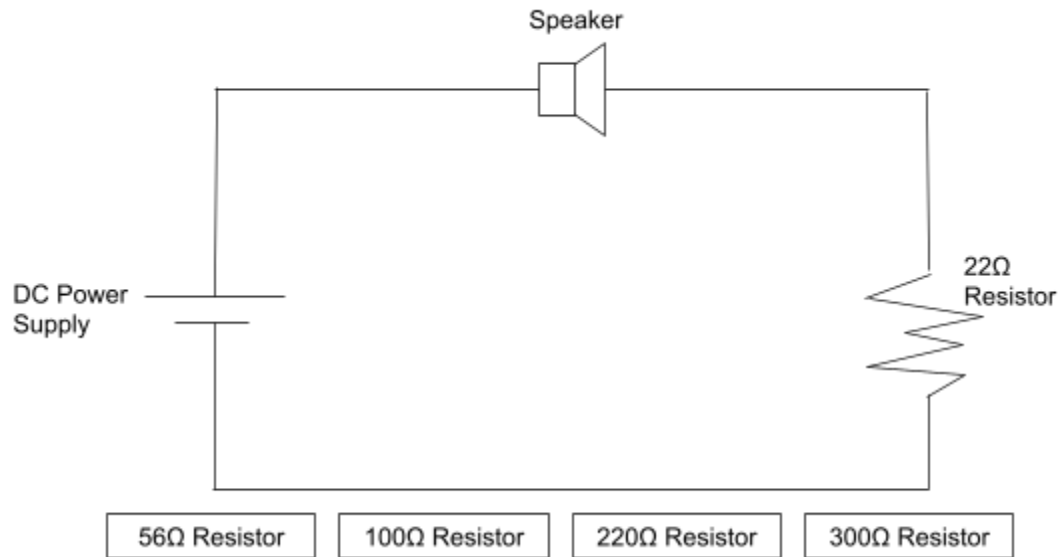


Diagram made by H.F. on January 6, 2025

Figure 2: A diagram of the lab set-up

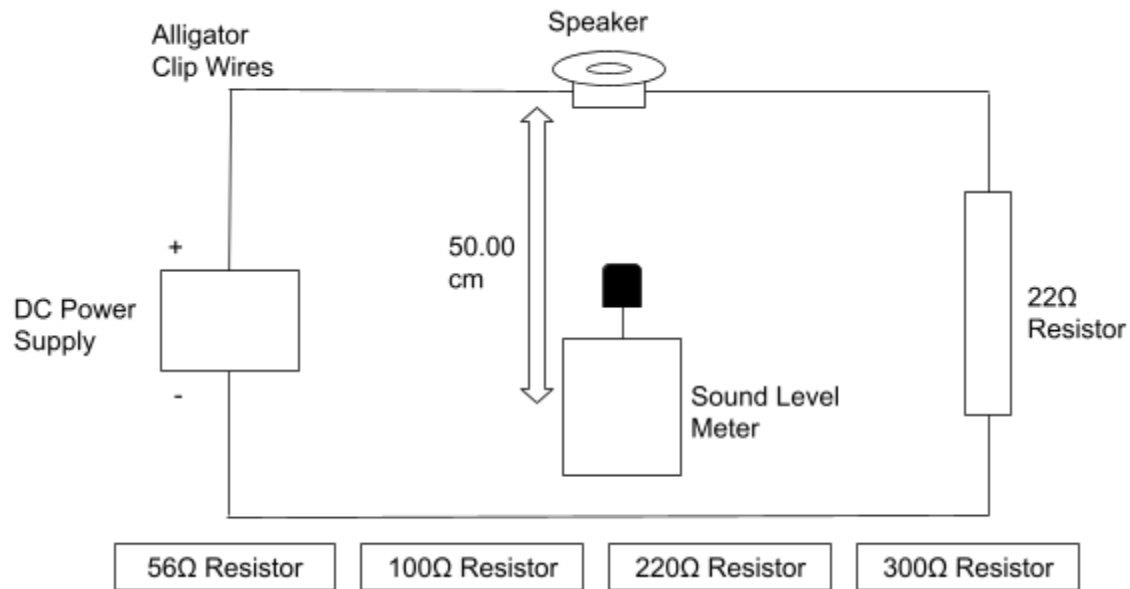


Diagram made by H.F. on January 22, 2025

The resistor's resistances selected in this experiment were chosen based on their availability and to represent a wide range of resistances.

Problem: How is the Sound Intensity, I (W/m^2) of Speakers Affected by Resistance, R (Ω) in their Circuits?

Hypothesis:

If a speaker and resistor are wired in series into a circuit with constant input voltage, then increasing the resistance of the resistor will decrease the sound level and sound intensity the speaker produces. As resistance in the resistor increases, it will follow Ohm's Law, given by $V = IR$ as an Ohmic resistor, resulting in a larger voltage drop across it as current is constant. Since input voltage to the circuit also remains the same, there will be less voltage across the speaker. For the same amount of current in the circuit, this will result in less power supplied to the speaker, producing lower sound levels. This is shown by the formula for electric power, $P = IV$, as electric power varies directly with voltage when the current is held constant, $P \propto V$. A decrease in power to the speaker will thus decrease its sound levels produced. As given by the rearranged formula for sound intensity, $I = 10^{\frac{L}{10}} \cdot 10^{-12}$, decreasing sound levels will also decrease sound intensity, and thus sound intensity will vary inversely to the resistance in the given circuit. Although a PA system is likely configured in a more complex circuit system, this relationship would still stand and cause the sound intensity produced by a speaker to decrease when more resistance is wired into the circuit, making the sound perceived by the human ear decrease.

Variables:

Manipulated variable: Resistance of resistor (in Ohms) connected in series to speaker

Responding variable: Sound level of the speaker (Decibels)

Controlled variables: Shown in Table 1 below

Table 1: The controlled variables present in this lab, why they must be controlled, and how they will be controlled under the premise of this lab.

Controlled Variable	Why It Must Be Controlled	How It Will Be Controlled
Input voltage of the circuit, V_i	From Ohm's Law, $V = IR$, changing the input voltage of the circuit will also change the current flowing through the circuit for a given resistance which will impact the sound level. Input voltage thus needs to be controlled to isolate the effect of changing circuit resistance.	The input voltage will be kept at a constant value throughout all the trials. This is done so by fixing the dial on the DC power supply to 2V.
Distance from speaker to sound level meter	As sound level also varies with distance from the source, the distance from the sound level meter to the source must be controlled to isolate the manipulated variable to the resistance. Sound intensity in particular decreases dramatically as distance increases so this variable must be kept constant.	The sound level meter will be a constant distance away from the speaker throughout all trials.
External noise	The sound level meter will measure all sound levels detected and additional noises from the speaker are going to be picked up as well. As the experiment doesn't rely on getting an absolute value for the sound level, it is important for the external noise to not fluctuate. External noise thus must be controlled to minimize the sound level being affected by sources other than the circuit.	The experiment will be conducted in a quiet environment, away from places with high noise output or varying levels of noise.
Atmospheric conditions of lab	How sound waves travel depends on its medium, and can be affected by conditions like air temperature. Thus, the atmospheric conditions of the lab must be held constant to isolate the effect of changing resistance.	The lab will be conducted in the same location throughout all trials.
Internal resistance of circuit components	As all the circuit components have internal resistances, the same components must be used within the circuit across trials to ensure that its effect on the total circuit resistance is kept constant.	Alligator clip wires of the same make and model will be used throughout all trials, along with the same DC power supply and speaker.

Materials

- 25W DC power supply

- 3 Alligator clip wires
- 1 Speaker
- 1 of each resistor in resistances of 22Ω , 56Ω , 100Ω , 220Ω , and 300Ω
- Sound level meter
- Tape
- 1 Metre Stick

Procedure

1. Gather all of the required materials.
2. Construct the circuit pictured in Fig. 1, ensuring that all of the wires are secured with alligator clips and not entangled together (if wires are tangling together, some tape can be used as wire management).
3. Ensure that the DC power supply is on the DC terminal, and turn the dial until the voltmeter reading reads 2.0V. The speaker should begin creating a sound.
4. Using the metre stick, measure a 50.00cm distance from the speaker, and place the centre of the sound level metre at that location. Place one piece of tape on each side of the meter to mark its place.
5. Turn the sound level meter on.
6. Wait until the sound level reading stabilizes on the sound level meter, then press the “Hold” button on it to freeze its reading at that point. Record the sound level measured along with any other qualitative observations.
7. Repeat the measurement in step 6 until 5 data points for the resistance have been collected.
8. Maintaining the construction and configuration of the circuit in step 2, replace the 22Ω resistor with resistors of 56Ω , 100Ω , 220Ω , and 300Ω resistances, and repeat steps 6-7.
9. Disconnect the DC power supply from its power source, and disconnect the rest of the wires from each other, putting them away neatly to store for future use.
10. Power off the sound level meter, and clean up the remaining circuit components.

Safety

- To avoid circuit components overheating, the power source will be unplugged when not in use to ensure that it is only plugged in for minimal amounts of time
- Physical contact with the circuit components will only be when necessary to minimize the risks of burning in case the circuit components heat up too quickly
- The maximum voltage (input voltage) of the circuit will be controlled at 2.0V to prevent potential electric shock hazards
- Conduct the lab in a place away from substances and items prone to creating electrical hazards, such as water

Ethics

- There are no ethical concerns in this lab

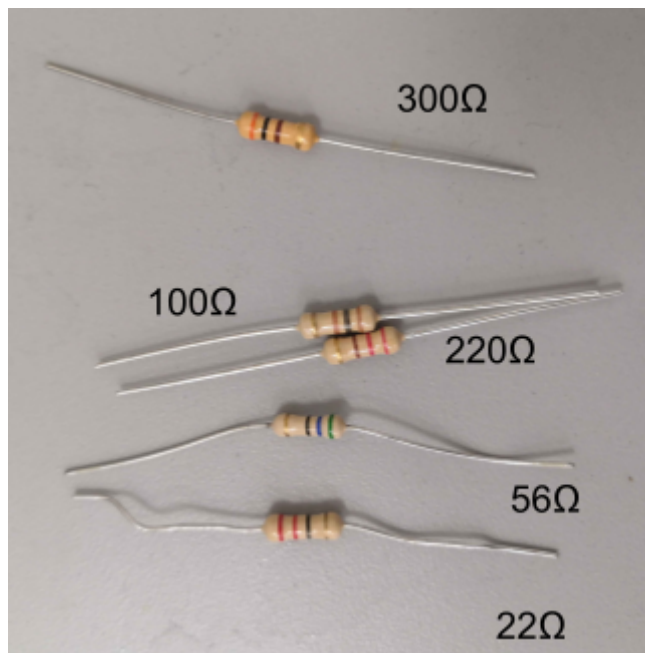
Environment

- Minimal amounts of tape will be used then discarded, though no other waste is produced
- All the other equipment in this lab will be reused for future labs to reduce the waste they produce
- The power supply will be unplugged when not in use
 - Overall, the power consumption in this lab will be negligible when compared to average household consumption and will not be a major environmental concern

Qualitative Observations

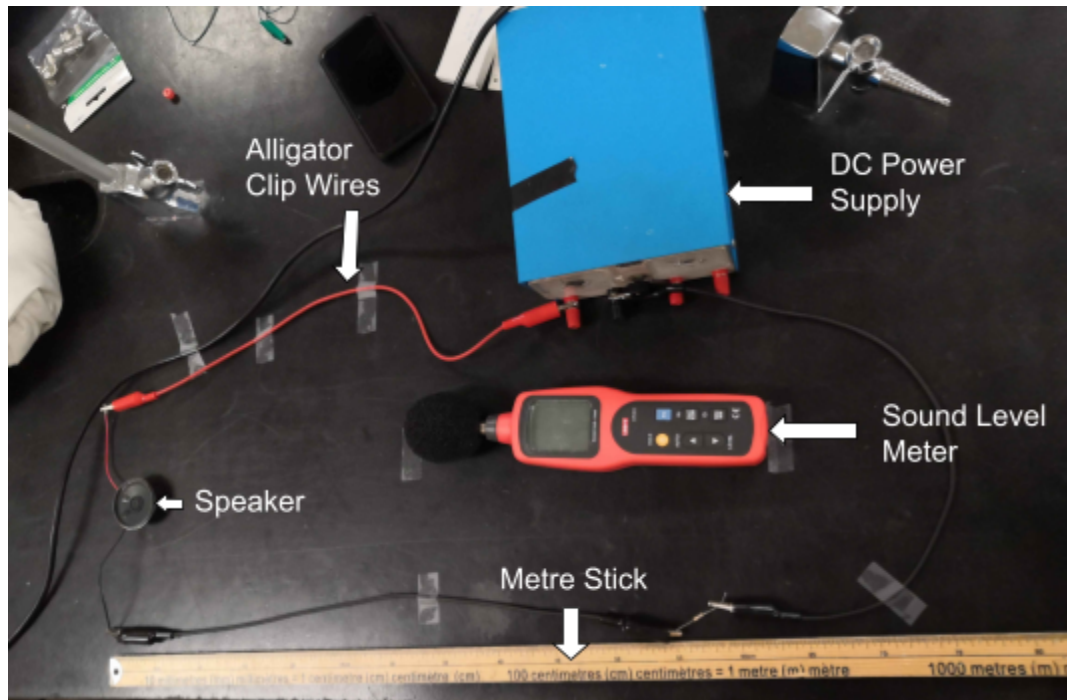
- Vibrations could be felt through wires near the speaker when it was playing
- Readings on the sound level meter never stabilized for more than one or two seconds due to small shuffles and other sound disruptions in the surrounding environment
- The sound the speaker produced audibly wavered from time to time
- A noticeable decrease in volume occurred as resistance increased
- The decibel meter was very sensitive: whispers, bumps, and shuffles that seemed audibly small in sound all caused a spike in the reading

Figure 3: The resistors used in this lab



Picture taken by H.F. on January 15, 2025

Figure 4: The experimental set-up and circuit configuration



Picture taken by H.F. on January 15, 2025

Tolerance of resistors used:

As every resistor had a gold band at the end (see Figure 2), their tolerances are each 5%.

Accuracy of the sound level meter:

The instruction manual attached gave an uncertainty value of $\pm 1.5\text{dB}$.

Table 2: The resistances of the resistors in the circuit and their corresponding sound levels of the speaker produced

	Sound level, L (± 1.5 dB)				
Resistance, R , (Ω), ($\pm 5\%$)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
22	69.9	70.4	70.2	70.5	70.0
56	63.9	63.9	64.7	64.0	65.6
100	59.9	60.2	59.6	59.7	59.6
220	54.6	54.0	54.8	54.3	54.8
300	52.0	52.6	52.8	52.5	52.4

Analysis

Table 3: Resistance of resistor in the circuit with absolute uncertainty in resistance, and the corresponding average sound level with absolute uncertainty in average sound level.

Resistance, R , (Ω)	Absolute Uncertainty in Resistance, ΔR (Ω)	Average Sound Level, L (dB)	Absolute Uncertainty in Average Sound Level, ΔL (dB)
22	1.1	70.2	0.3
56	3	64.4	0.9
100	5	59.8	0.3
220	11	54.5	0.4
300	15	52.5	0.4

Sample calculations for the data in row 2 of the data in Table 3:

Absolute uncertainty in resistance:

$$\varepsilon R = \frac{\Delta R}{R}$$

$$\Delta R = \varepsilon R * R = 5\% * 56\Omega = 3\Omega$$

Average sound level:

$$L_{avg} = \frac{L_1 + L_2 + L_3 + L_4 + L_5}{5} = \frac{63.9dB + 63.9dB + 64.7dB + 64.0dB + 65.6dB}{5} = 64.4dB$$

Absolute uncertainty in average sound level:

$$\Delta L = \frac{\text{Largest value} - \text{smallest value}}{2} = \frac{65.6dB - 63.9dB}{2} = 0.9dB$$

Table 4: Resistance of the resistors in the circuit and its corresponding average sound intensity and absolute uncertainty in average sound intensity.

Resistance, R , (Ω)	Absolute Uncertainty in Resistance, ΔR (Ω)	Average Sound Intensity, I (W/m^2)	Absolute Uncertainty in Average Sound Intensity, ΔI (W/m^2)
22	1.1	$1.1 * 10^{-5}$	$7.2 * 10^{-7}$
56	3	$2.8 * 10^{-6}$	$5.7 * 10^{-7}$
100	5	$9.6 * 10^{-7}$	$6.6 * 10^{-8}$
220	11	$2.8 * 10^{-7}$	$2.6 * 10^{-8}$
300	15	$1.8 * 10^{-7}$	$1.6 * 10^{-8}$

Sample calculations for the data in row 2 of the data in Table 4:

Average sound intensity:

$$I_{avg} = 10^{\frac{L_{avg}}{10}} \cdot I_0 = 10^{\frac{64.4dB}{10}} \cdot I_0$$

$$I_{avg} = 2.8 * 10^{-6} W/m^2$$

Absolute uncertainty in average sound intensity:

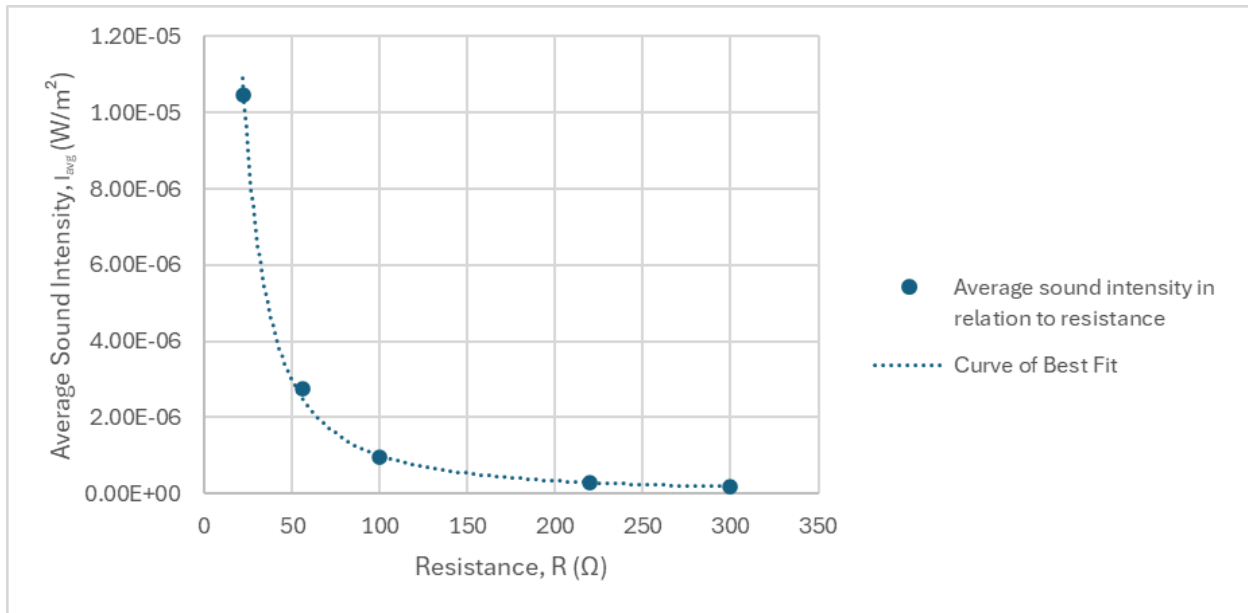
$$\Delta I_{avg} = \frac{\text{Maximum value} - \text{minimum value}}{2}$$

$$\text{Maximum value} = 10^{\frac{(64.4 + 0.9)dB}{10}} \cdot I_0 = 3.388.. W/m^2$$

$$\text{Minimum value} = 10^{\frac{(64.4 - 0.9)dB}{10}} \cdot I_0 = 2.238.. W/m^2$$

$$\Delta I_{avg} = \frac{3.388.. W/m^2 - 2.238.. W/m^2}{2} = 5.7 * 10^{-7} W/m^2$$

Figure 5: Graph of the average sound intensity as a function of resistance



As the trend shown above is a curve, the data must be adjusted to display a linear relationship. The curve of the graph displays a curve similar to an inverse curve, meaning that the variables seem to be inversely related. This can be demonstrated by: $I_{avg} \propto \frac{1}{R}$. To manipulate the data to display a linear relationship, taking the inverse of one of the variables, resistance, and graphing its relationship with average sound intensity should produce a straightened trendline.

Table 5: The inverse of resistance and its absolute uncertainty, and the corresponding average sound intensity and absolute uncertainty in average sound intensity.

Inverse of Resistance, R^{-1} , (Ω^{-1})	Absolute Uncertainty in the Inverse of Resistance, ΔR^{-1} (Ω^{-1})	Average Sound Intensity, I (W/m^2)	Absolute Uncertainty in Average Sound Intensity, ΔI (W/m^2)
$4.5 * 10^{-2}$	$2 * 10^{-3}$	$1.1 * 10^{-5}$	$7.2 * 10^{-7}$
$1.8 * 10^{-2}$	$1 * 10^{-3}$	$2.7 * 10^{-6}$	$5.7 * 10^{-7}$
$1.0 * 10^{-2}$	$1 * 10^{-3}$	$9.6 * 10^{-7}$	$6.6 * 10^{-8}$
$4.5 * 10^{-3}$	$2 * 10^{-4}$	$2.8 * 10^{-7}$	$2.6 * 10^{-8}$
$3.3 * 10^{-3}$	$2 * 10^{-4}$	$1.8 * 10^{-7}$	$1.6 * 10^{-8}$

Sample calculations for data in row 2 of the data for Table 5:

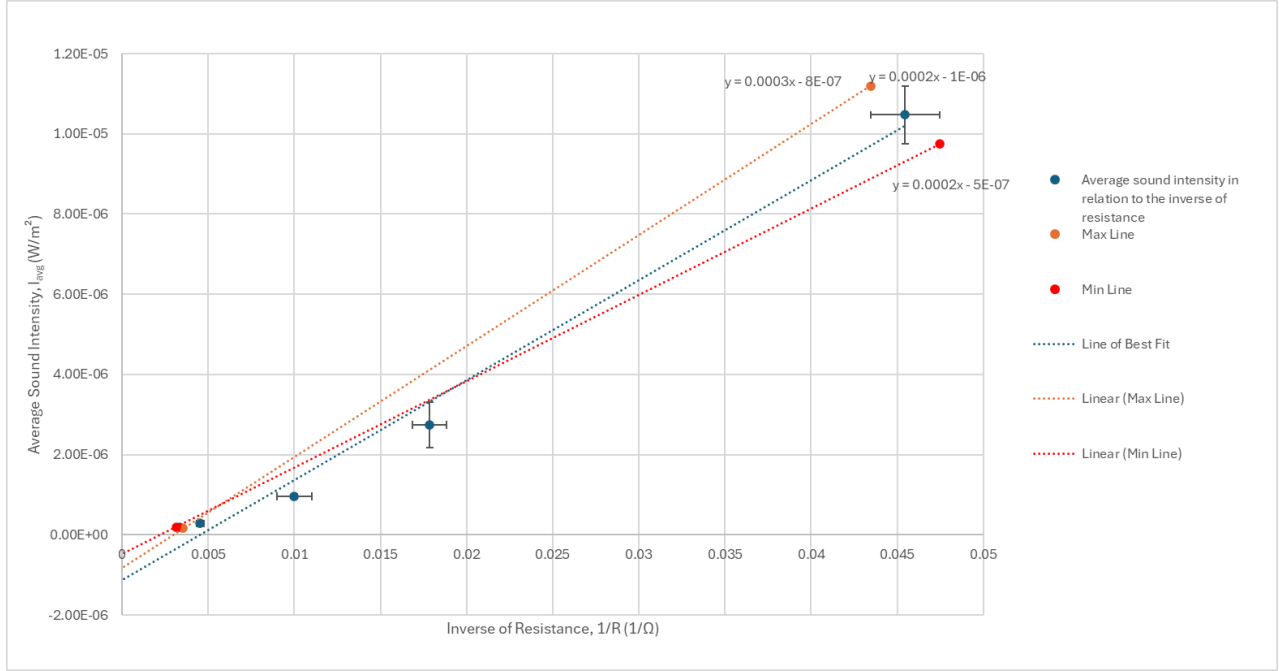
Inverse of resistance:

$$R^{-1} = 56\Omega^{-1} = 1.8 * 10^{-2} \Omega^{-1}$$

Absolute uncertainty in the inverse of resistance:

$$\Delta R^{-1} = \frac{\text{Maximum value} - \text{minimum value}}{2} = \frac{((56-3)\Omega)^{-1} - ((56+3)\Omega)^{-1}}{2} = 1 * 10^{-3} \Omega^{-1}$$

Figure 6: Graph of the curve-straightened relationship, average sound intensity as a function of the inverse of resistance.



The uncertainty values for some points are too small to be seen on the graph, or are covered by points from the maximum and minimum lines. One of the data points is also completely covered by points from the maximum and minimum lines near the bottom of the graph. For these values, refer to Table 5.

Graphical Analysis for Figure 6:

The equation of a line can be written in the form $y = mx + b$, where m is the slope of the line and b is the y-intercept of the line.

From the equations in the graph:

$$m = 3 * 10^{-4} \frac{W}{m^2 \Omega}$$

$$m_{min} = 2 * 10^{-4} \frac{W}{m^2 \Omega}$$

$$m_{max} = 3 * 10^{-4} \frac{W}{m^2 \Omega}$$

$$\Delta m = \frac{m_{max} - m_{min}}{2} = 4 * 10^{-5} \frac{W}{m^2 \Omega}$$

$$b = -1 * 10^{-6} \frac{W}{m^2}$$

$$b_{min} = -5 * 10^{-7} \frac{W}{m^2}$$

$$b_{max} = -8 * 10^{-7} \frac{W}{m^2}$$

$$\Delta b = \frac{b_{max} - b_{min}}{2} = 6 * 10^{-7} \frac{W}{m^2}$$

Meaningful Equation for the Line of Best Fit in Figure 5:

$$I_{avg} = (3 * 10^{-4} \frac{W}{m^2 \Omega} \pm 4 * 10^{-5} \frac{W}{m^2 \Omega}) R^{-1} - 1 * 10^{-6} \frac{W}{m^2} \pm 6 * 10^{-7} \frac{W}{m^2}$$

Evaluation of the relationship:

As seen from the graph in Figure 4, a relationship between the two variables, average sound intensity of the speaker and the resistance in the circuit does indeed exist. The curve of the graph is exponentially decaying, demonstrating an inverse relationship between the two variables. After manipulating the data to straighten the curve of the graph, we obtain the graph in Figure 5. This linear trendline produced by the graph of the average sound intensity as a function of the inverse of resistance demonstrates that the two variables indeed follow an inverse relationship with one another, proving the hypothesis of the experiment correct and the goal of the experiment— to find a relationship between the two variables has been achieved.

Sources of Error

Table 6: Sources of error occurring in this experiment, their classifications, impacts on the results of the lab, and improvements for the future.

Error	Classification	Impact	Improvement
Background noises are inconsistent	Random Major Error	As the sound level meter was quite sensitive to any movement or disruptions that caused sounds, its readings would always fluctuate between values due to inconsistencies in background noise.	Conduct the experiment in a completely soundproof environment.
Large gap in data	Systematic Major Error	As the resistors used were chosen for their availability, the resistors did not increase by a constant amount. It thus left larger ranges of resistance values that were not tested, creating larger gaps in the graph.	Obtain resistors that are closer in resistance to each other, in order to obtain results that are closer together and more accurate.
Speaker sound was inconsistent during trials.	Random Minor Error	As the sound output from the speaker wavered from time to time, it caused the sound level detected on the meter to flicker between multiple different sound levels, producing an unsteady reading.	Increase input voltage of the circuit slightly (within safe range of the power supply), or obtain a speaker with clear and stable sound output.
Internal resistance in wires and circuit components	Random Minor Error	The wires used in the experiment and the speaker itself all have internal resistance, which was not accounted for in the data calculations. Thus, the absolute values of resistance should be more than what is recorded.	Measure the internal resistance of all the circuit components and take them into account during analysis.

Conclusion

The experiment conducted proved the hypothesis to be true: that a relationship between the resistance in a circuit and a speaker's sound intensity existed, and was in an inverse relationship. From Figures 4 and 5, the trendlines on the graph show that an inverse relationship between the two variables resulted in the volume changes observed during the lab. Although only five trials were conducted in the lab, the relationship shown in the graph is still visibly clear. In the equation calculated from the trendlines on the graph, the relationship between the variables, the inverse of resistance and sound intensity is quantitatively given. For every increment of the inverse of resistance, the sound intensity increases by a factor of $3 * 10^{-4} \frac{W}{m^2 \Omega}$ by the inverse of resistance, subtracted by $1 * 10^{-6} \frac{W}{m^2}$. The y-intercepts of the line of best fit, maximum line, and minimum line were all relatively close to the origin, and the overall uncertainty in the data is

relatively small, as the order of magnitude of the uncertainties were significantly smaller than the values of the slope of the line of best fit. In particular, the y-intercept represents the sound intensity when the inverse resistance is zero (making the resistance infinitely large). This value is expected to be zero as a circuit with infinitely high resistance will not result in sound being produced. In this lab, it was found to be a small negative number, which can be attributed to errors. To further confirm the relationship between the two variables, more resistor values could be used, especially ones that are in smaller increments compared to the ones selected in the experiment. As seen in Table 1, the smallest grade resistor of 22Ω produced sound levels hovering around 70dB, which is on the boundary of safe hearing for human ears. As resistance increased, it caused a decrease in the sound levels and intensities of the speaker, thus bringing it well within a safe hearing range. Thus, increasing resistance in a circuit will allow for speakers in PA systems to be within safety regulations, and is thus one of the solutions to reducing sound levels to being safe like I initially pondered about.

Extension

To further improve the lab and increase its similarities to modeling a public address system, multiple speakers wired to amplifiers could be used. As PA systems are used throughout schools and need to reach classrooms, hallways, and other facilities inside of the building, amplifiers are necessary to project the sound over large areas. Thus, to further improve and replicate these systems, the effect of resistance on speakers with the assistance of amplifiers could be used to further investigate the relationship. Additionally, the distance from the observer to the speakers could be accounted for in the analysis, as the speakers in a building are typically attached to the roof, and thus have some distance to travel before reaching the ears of people below.

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